

Project Submission Requirements and Guides

I. Preface

1. During the semester, you will be implementing about 8 programming assignments (projects).
2. For each project, you will be given the project specs which includes: Programming language (C++ or Java), due date, the functional specs, input/output specs, data structure, algorithm steps for the main and other functions. See an example.
3. Project submission will be via email; every email submission requires "**soft copy**" and "**hard copy**" attachments.
4. In section II, you will learn how to name your "**soft copy**" and "**hard copy**" of your project.
(Note: your submitted soft copy will be unzipped, then compiled and executed by a script; failing to follow the naming convention means your project will not be recognized; hence, you will receive a zero (0) grade.)
5. In section III, you will learn what is the "**soft copy**" of your project and how to create it.
(If you created multiple .cpp or .java files in a project, you must put all of them into a single .cpp or .java file.)
6. In section IV, you will learn what is the "**hard copy**" of your project and how to create it.
7. In section V, you will learn how to submit your project via email.
8. In section VI, you will learn how to download C++ and Java compilers and the instructions of the compilation and execution for your C++ code and Java codes.
9. **Before you implement your first C++ project and your first Java project, you must watch the project submission guides via YouTube given in section VII. Screen recording requirement is given in section VIII.**

II. Naming Conventions for soft copy and hard copy

A) The "soft copy" of a project is a zip file of .cpp or .java of your project's source code.

The naming convention of a "soft copy" includes three parts:

- i) The first part is your last name with your first name initial;
- ii) the second part is project #;
- iii) the third part is Main.

These three parts are concatenated with underscores. For example, if your first name is John and last name is Smith; then, the first part is SmithJ, the second part is Project#, the third part is Main. Below are examples for project#: (# means 1 or 2 or 3 or, ..., 8; for project1, project2, project 3, ..., project8).

For C++:

SmithJ_Project#_Main.cpp // this is the name of your .cpp file for project#

For Java:

SmithJ_Project#_Main.java // this is the name of your .java file for project#

B) The "hard copy" of a project is a word file that includes: a cover page, the project source code, and the project outputs.

The naming convention of a "hard copy" will be (for C++ and Java):

SmithJ_Project#_DOC.docx // where # is the number of the project, i.e., 0A, 0B, 1, 2, 3, ...

III. Soft Copy (a zip file) construction steps

Step 1: Zip your .cpp/.java file. (**Zip only the source code!** Nothing else! Do not include your executable!)

After zip, the name of zipped file for SmithJ_Project#_Main.cpp will be: SmithJ_Project#_Main.cpp.zip

After zip, the name of zipped file for SmithJ_Project#_Main.java will be: SmithJ_Project#_Main.java.zip

Step 2: Change the name of your .zip file as below:

For C++:

change SmithJ_Project#_Main.cpp.zip → SmithJ_Project#_CPP.zip

For Java:

change SmithJ_Project#_Main.java.zip → SmithJ_Project#_JAVA.zip

**** You will attach the .zip file ("soft copy") in the attachment of your project submission Gmail!**

IV. Hard Copy (a word file) construction steps

Step 1: Open a word document (use the naming convention for hard copy, such as “SmithJ_Project#_DOC”)

Step 2: Include the following in the word file:

- i) Cover Page: (may be 1 or 2 pages)
your class name (323 or 700); your name; project #; project name; C++ or Java, and due date and submission date; then the algorithm steps on the main (..) function. See example given.
- ii) Illustrations: (only if required in the project specs.)
- iii) Source code: Copy and paste your project source code. (Do not screenshot any part of your code!!!)
- iv) Program output: Copy and paste the "output" files your program produces.

*** The name of the hard copy should be:

SmithJ_Project#_DOC.docx // The same for both C++ project and Java Project.)

*** **You will attach the .docx file (hard copy) in the attachment of your project submission Gmail!**

V. Email submission instructions

Submit your project via email, **one project per Gmail (do not use QC gmail!!)** Below is the template of your email when you submit a project.

To: your TA's email address (given in your class syllabus)

cc: Tsaiyun.Phillips@qc.cuny.edu //Dr. Phillips' email address

Subject: (class name) firstName lastName <Project #: Project Name (C++) or (Java)>

For example, the email subject for your project 0A and project 0B

For Java project (circle your class name)

(323 or 700) John Smith <Project 0A: Java I/O & project submission exercise (Java)>

For C++ project (circle your class name)

(323 or 700) John Smith <Project 0B: C++ I/O & project submission exercise (C++)>

*** Please note, your email subject line must be written exactly as given in the above format. Otherwise, your project submission will not be recognized by the script; hence, your grade will be “non-submission”.

Attachments: attach both of soft copy (zip file) and hard copy (docx file) in the attachments.

For example,

Attachments: "SmithJ_Project1_DOC.docx" and "SmithJ_Project1_CPP.zip" are considered valid submission.

Email body: The body of your submission email must have the following contents. Copy and paste the following and replace the blanks with appropriate answers, i.e., yes or no.

Student: _____

Submitting Project _____

- 1) Does your code pass compilation with g++ command (for C++), or javac command (for Java)? _____.
- 2) Does your project execute? _____.
- 3) Does your project produce results? _____.
- 4) Did you write your program by yourself? _____.
- 5) How many hours did you spend on this project? _____.

Place your screen recording here.

VI. Installation & Compilation & Execution:

A) Installing C++ compiler

For Windows-

▶ How to install MinGW C/C++ Compiler on Windows 11 - GCC G++ Installation Tutorial

For Mac-

Preferred installation:

Most MacOS already has g++ installed. If not, you can download the latest version of XCode.

<https://developer.apple.com/xcode/>

Alternative installation:

▶ Install GCC/G++ on MacOS arm64 M1 Mac

B) Installing Java JDK

For this course, any java version 8+ is allowed.

For Windows-

▶ How to Install Java JDK 17 on Windows 10

For Mac-

▶ How To Download & Install Java 17 on Mac

C) Compilation & Execution:

- Before you submit your project, make sure your .zip file has the correct name as given in section II.
- The .zip file of your source code will be unzipped, compiled and executed.
- Assuming "Project1" accepts 3 parameters; "data1.txt", "output.txt", and "logFile.txt".
 - **Note: in your C++ source code, the 3 parameters are: argv [1], argv [2], and argv [3].
in your Java source code, the 3 parameters are: args [0], args [1], and args [2].
- Make sure the 3 .txt files are in your current directory.

Your C++ source code (unzipped) needed to (will be) compiled and executed in the following way:

C++ compilation command (for project1, for example):

```
g++ -std=c++11 SmithJ_Project_Main.cpp -o SmithJ_Project1_Main
```

unzipped .cpp file

executable

C++ Execution command:

```
./SmithJ_Project1_Main data1.txt output.txt logFile.txt
```

executable

argv [1]

argv [2]

argv [3]

Your Java source code needed to (will be) compiled and executed in the following way:

Java Compilation command:

```
javac SmithJ_Project1_Main.java
```

un-zipped .java file

Java Execution command:

```
java SmithJ_Project1_Main data1.txt output.txt logFile.txt
```

executable

args [0]

args [1]

args [2]

Note: double check that the project source code you submitted can be compiled with the above commands. There are some IDE's that ignore compilation errors and let you execute your code within the IDE. If your source code does not compile with those commands, then it will not compile in any machine other than your laptop; hence, you will receive a negative grade.

VII. Project submission guides via YouTube

Java project submission guide in YouTube video:

<https://www.youtube.com/watch?v=271QaOkY54g>

C++ project submission guide in YouTube video:

<https://www.youtube.com/watch?v=kv5aQSFm7Js>

VIII. Screen Recording Requirements:

A complete screen recording of running the compiler and execute commands of C++ or Java via the terminal. Your screen recording in essence should look like below:

For C++: from 1:10 to 2:55 min mark.

<https://www.youtube.com/watch?v=kv5aQSFm7Js>

For Java: from 2:35 to 3:25 min mark.

<https://www.youtube.com/watch?v=271QaOkY54g>

Once the video is recorded, upload to YouTube with an unlisted link. When submitting your project, include the video link in the email body.